Nicole M. Ford

UAT Keywords: Active Galactic Nuclei; Galactic Center; Supermassive black holes

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Interests

I use ground- and space-based telescopes to study weakly accreting supermassive black holes and their fuel reservoirs, including in our Galactic Center.

Experience

NSERC/FRQNT Graduate Fellow - Active Galactic Nuclei 2023 – 2027
Trottier Space Institute (TSI), *Advisor*: Daryl Haggard

Graduate Research Assistant - Neutron Star Mergers 2021 – 2023
Trottier Space Institute (TSI), *Advisors*: Daryl Haggard & John Ruan

DOE SULI Researcher - Computational Astrophysics 2020 – 2021
Lawrence Berkeley National Laboratory, *Advisor*: Dr. Ann Almgren

Clare Boothe Luce Scholar - Star Forming Galaxies 2019 – 2020
Williams College, *Advisor*: Professor Anne Jaskot

CERN Researcher - Solar Cosmic Rays Winter – Summer 2019
University of Geneva & Boston University, *Advisor*: Dr. Maura Graziani

Science Research Fellow - Star Forming Galaxies Spring – Summer 2018
University of Massachusetts, Amherst, *Advisor*: Professor Anne Jaskot

NSF REU Fellow - Exoplanet Transits Spring – Summer 2017
Wellesley College, *Advisor*: Professor Kim McLeod

Education

McGill University, Montréal, QC
Ph.D. in Physics, *Advisor*: Daryl Haggard expected 2027
M.Sc. in Physics, *Advisor*: Daryl Haggard & John Ruan 2023
Thesis: "KilonovAE: Exploring Kilonova Spectral Features with Autoencoders"

Williams College, Williamstown, MA 2016 – 2020
B.A. with Highest Honors in Astrophysics & Studio Art, *Advisor*: Anne Jaskot
Thesis: "Optical Properties of Low-Redshift Star-Forming Galaxies with Potential Lyman Continuum Escape"

Honors & Awards

NSERC Doctoral Scholarship, $120,000 over 36 months 2026
Awarded by the Natural Sciences and Engineering Research Council of Canada.

FRQNT Doctoral Research Award, $100,000 over 48 months 2023
Awarded by Fonds de recherche du Québec – Nature et technologies. Ranked 1st of 15 candidates in the Astronomy & Astrophysics category.

SciArt200 "Traditional Media" Prize, $200 2022
Awarded by McGill University for their Bicentennial Art & Science Exhibition

Trottier Space Institute Fellowship, $3000 over 36 months 2021-2024
Awarded by McGill University

Chambliss Astronomy Achievement Award 2020
Awarded by the AAS in the "Undergraduate" category

Clare Boothe Luce Scholar, Williams College 2018

PUBLICATIONS

Summary: Three first-author refereed publications; one co-first authored with a supervised student (in prep); fifteen as co-author, including one led by a graduate student mentee. Additional publications can be found on NASA ADS. Total citations: 700+. h-index: 9.
NB: * marks an undergraduate or graduate student under my supervision.

First Author:

[4] **Ford, N. M.** & Pereira, O.*, et al., "A TESS Look at Rapid Optical Variability in Faint and Luminous AGNs", ApJ, *in prep* .

[3] **Ford, N. M.**, Balakrishnan, M., et al., "Ionization Structure and Metal Enrichment of the Galactic Center Minispiral Observed with JWST", ApJ, *in press*.

[2] **Ford, N. M.**, Nowak, M., et al., "Tracking X-ray Variability in Next Generation EHT LLAGN Targets", ApJ 981.2 (2025).

[1] **Ford, N. M.**, Vieira, N., "KilonovAE: Exploring Kilonova Spectral Features with Autoencoders", ApJ 961.1 (2024).

Co-Author (Selected Works):

[15] Sumners, Z.*, **Ford, N. M.**, et al., "The Demographics of Sagittarius A* X-ray Flares over 25 Years with Chandra", ApJ (*in press*).

[14] Balakrishnan, M., et al., (*incl.* **Ford, N.M.**), "Infrared Line Diagnostics Fail to Constrain Sgr A*'s UV Output", ApJ 1008.1 (2026).

[13] Michail, J. et al. (*incl.* **Ford, N. M.**), "Detection and Evolution of Linear Polarization of the Galactic Center Transient MAXI J1744-294", ApJ 1003.1 (2026).

[12] Michail, J., et al. (*incl.* **Ford, N. M.**), "First Mid-infrared Detection and Modeling of a Flare from Sgr A*. II. Submillimeter Polarimetry and Spectral Energy Distribution", ApJ 997.2 (2026).

[11] Roychowdhury, T., et al. (*incl.* **Ford, N. M.**), "Photometric Constraints on Intermediate-mass Black Holes in the Galactic Centre", PASP 127.11 (2025).

[10] von Fellenberg, S. D., et al. (*incl.* **Ford, N. M.**), "Mid-infrared extinction toward the Galactic center", ApJ 995.2 (2025).

[9] von Fellenberg, S. D., et al. (*incl.* **Ford, N. M.**), "First Mid-infrared Detection and Modeling of a Flare from Sgr A*", ApJL 979.1 (2025).

[8] Algaba, J. C. et al. (*incl.* **Ford, N. M.**), "Broadband Multi-wavelength Properties of M87 during the 2018 EHT Campaign including a Very High Energy Flaring Episode", A&A 692, A140 (2024).

[7] Vieira, N., Ruan, J. J., Haggard, D., **Ford, N. M.**, et al., Spectroscopic r-Process Abundance Retrieval for Kilonovae II: Lanthanides in the Inferred Abundance Patterns of Multi-Component Ejecta from the GW170817 Kilonova, ApJ 962.1 (2024).

[6] Vieira, N., Ruan, J. J., Haggard, D., **Ford, N. M.**, et al., "Spectroscopic r-Process Abundance Retrieval for Kilonovae I: The Inferred Abundance Pattern of Early Emission from GW170817", ApJ 944.2 (2023).

[5] Flury, S., et al. (*incl.* **Ford, N. M.**), "The Low-Redshift Lyman Continuum Survey II: New Insights into LyC Diagnostics", ApJ 930.2 (2022).

[4] Flury, S., et al. (*incl.* **Ford, N. M.**), "The Low-redshift Lyman Continuum Survey. I. New, Diverse Local Lyman Continuum Emitters", ApJS 260.1 (2022).

[3] Richers, S., Willcox, D. E., **Ford, N. M.**, "Neutrino fast flavor instability in three dimensions", PRD 104.10 (2021).

[2] Richers, S., Willcox, D. E., **Ford, N. M.**, and Myers, A., "Particle-in-Cell Simulation of the Neutrino Fast Flavor Instability", PRD 103.8 (2021).

[1] Harpole, A., **Ford, N. M.**, et al., "Dynamics of Laterally Propagating Flames in X-ray Bursts. II. Realistic Burning & Rotation", ApJ 912.36 (2021).

Proceedings & White Papers:

[2] Turriziani, S. et al. (*incl.* **Ford, N. M.**), "Advancing Fundamental Physics and Cosmology with high-resolution X-ray imaging" NASA PhysPAG High-Resolution X-ray Imaging White Paper (2026).

[1] Nair, D. et al. (*incl.* **Ford, N. M.**), "Demographics of black holes at $< 100R_g$ scales: accretion flows, jets, and shadows", Proceedings of the 16th EVN Symposium (2024).

Astronomer's Telegrams:

[1] Michail, J. et al. (*incl.* **Ford, N. M.**), "Quick Radio Brightening of Galactic Center Transient MAXI J1744-294 at 33 and 43 GHz", 2025, ATEL 17174.

Invited & Contributed Talks

Summary: three invited talks; nine contributed talks; five posters

Conferences & Workshops:

Chandra's Dynamic Universe, Cambridge, MA	2026
Black Holes and Blue Notes, Montréal, Canada	2026

(Invited) EHT Collaboration Virtual Winter Meeting, Online 2025
EHT Collaboration Summer Meeting, Berlin, Germany 2025
ISSI Workshop, Bern, Switzerland 2025
EHT Collaboration Virtual Winter Meeting, Online 2024
(Session Chair) Joint NICER/IXPE Workshop, Washington, D.C. 2024
EHT Collaboration Summer Meeting, Mexico City, Mexico 2024
CRAQ Annual Meeting, Lac-à-l'Eau-Claire, QC 2024
EHT Collaboration Virtual Winter Meeting, Online 2023

Institutional Talks:

(**Invited**) Harvard Black Hole Initiative Group Seminar, Cambridge, MA 2025
(**Invited**) Canadian Institute for Theoretical Astrophysics, Toronto, ON 2025

Posters:

CASCA 2026, Montréal, Canada 2026
IAU Symposium 405 – Galactic Center, Brno, Czechia 2026
AAS HEAD 21, Horseshoe Bay, TX 2024
AAS 241, Seattle, WA 2021
(Chambliss Winner) AAS 235, Honolulu, HI 2020

OBSERVING ALLOCATIONS

Summary: 45+ hrs awarded as PI; hundreds of hours awarded as Co-I

Principal Investigator:

Chandra DDT (40 ks) 2025
"Chandra/Event Horizon Telescope Study of Messier 84: Jets and Accretion Inflow" (PI: **N. M. Ford**)

Swift TOO (10 ks) 2025
"Event Horizon Telescope Studies of Low Luminosity AGNs: Jets and Accretion in NGC 4261" (PI: **N. M. Ford**)

NuSTAR Cycle 10 (50 ks) + NICER (42 ks) 2024
"NuSTAR/NICER/Event Horizon Telescope Studies of Low Luminosity AGNs: Jets and Accretion" (PI: **N. M. Ford**)

Gemini North 2023A (7.8 hr) 2023
"Tick Tock: A Spectroscopic Investigation into an Imminently Merging Supermassive Black Hole Binary Candidate" (PI: **N. M. Ford**)

Co-Investigator:

Chandra Cycle 28 (100 ks) + JWST (36 hrs) 2026
"Probing Particle Injection in Sgr A*'s Flares with Joint Chandra X-ray and

JWST MIR Observations" (PI: J. Michail)

Chandra Cycle 28 (470 ks) + HST (33 hrs) + VLA (31 hrs) 2026
"Community Discovery Program: Panchromatic CXO/VLA/HST Observations of NS Mergers" (Co-PIs: S. Campana, W.-f. Fong, D. Haggard, R. Margutti)

ALMA DDT (20.5 hrs) 2025
"Full-Stokes Submillimeter Coverage during a JWST / Chandra-led Multi-wavelength Campaign of Sgr A*" (PI: J. Michail)

VLA 2025b (28 hrs) 2025
"VLA Observations of Sgr A* in Support of Possible SMA, MIRI, Chandra Monitoring" (PI: J. Michail)

JWST Cycle 4 (34.4 hrs) + Chandra (100 ks) 2025
"A Joint Mid-IR and X-ray Investigation of the Physics Driving Sgr A*'s Flares" (PI: J. Hora)

GMVA+APEX Session I (3.4 hrs) 2025
"Collimation and acceleration of the M84 jet: synergy with EHT" (PI: D. Nair)

EVN+eMERLIN Session I (9 hrs) 2025
"Collimation and acceleration of the M84 jet: synergy with EVN+e-MERLIN and EHT" (PI: D. Nair)

Chandra Cycle 26 & 27 (470 ks) + HST (33 hrs) + VLA (33 hrs) 2025
"Community Discovery Program: CXO/ VLA/ HST observations of GW-detected compact mergers in O4" (Co-PIs: D. Haggard, R. Margutti. W. F. Fong, S. Campana)

ATNF Long Baseline Array Semester 2025APR (24 hrs) 2025
"The collimation and acceleration of radio jets: Synergy with the EHT" (PI: D. Kim)

VLA DDT (7 hrs) 2024
"Concurrent Radio Coverage of 6 September 2024 Multiwavelength Campaign of Sgr A*" (PI: J. Michail)

ALMA Cycle 11 (8 hrs) 2024
"Imaging M84 at $<$ 50 gravitational radii: jets and accretion inflow" (PI: N. Nagar)

JWST Cycle 2 (30 hrs) + Chandra (75 ks) 2024
"Sgr A* as Particle Accelerator: What Drives the Black Hole's Variable IR and X-ray Emission?" (Co-PIs: J. Hora, D. Haggard, G. Witzel)

COMPUTING ALLOCATIONS

Senior Investigator:

NERSC Allocation, (18 M MPP hours) 2021
"Neutrino Flavor Transformation in Neutron Star Mergers"

BRIDGES/2 Allocation (4000 GPU hours) 2021
"Neutrino Flavor Instabilities in Neutron Star Mergers"

Co-Investigator:

NERSC Allocation (30 M MPP hours) 2021
"3D studies of white dwarfs, massive stars, and neutron star systems"

Teaching & Mentorship Experience

Graduate Mentees:

Nora Salem, McGill University (MSc) Fall 2026 – present

Zach Sumners, McGill University (MSc) Summer 2024 – Fall 2026
Thesis: "Disentangling Observations of the Galactic Center with Score-Based Diffusion Models"

Undergraduate Mentees:

Olivia Pereira, McGill University (U3) Spring 2025 – Summer 2025

Chaitanya Kumar, University of Toronto (U3) Spring 2024 – Summer 2024

Marissa Lindon, McGill University (U3) Winter 2023 – Summer 2023

Charlotte Garcia, McGill University (U3) Winter 2023 – Spring 2023

Teaching Assistantships:

Department of Physics, McGill University Fall 2021
Supervisor: Prof. Katelin Schutz

Hopkins Observatory, Williams College 2017 – 2020
Supervisors: Dr. Steven Souza & Dr. Kevin Flaherty

Academic Service

Early Career Council Americas Representative, EHT 2025 – present

Reviewer, The Astrophysical Journal 2025 – present

Executive Secretary, NASA GO Review Panel 2024 – present

Community Engagement

Public Talks:

Astronomy On Tap, Montreal, QC 2023

Trottier Space Institute Lunch Talk, Montreal, QC 2022

McGill Bicentennial Space Week, Montreal, QC 2022

Outreach & Inreach:

Astronomy on Tap Organizer, McGill / University of Montreal 2024 – present

Scientista McGill Mentor, McGill University 2023 – present

Science in Space Mentor, TSI and Dell Technologies 2022 – present

Observatory Guide, McGill Observatory 2022 – 2023

Graduate Seminar Coordinator, TSI 2022 – 2024

AstroMcGill Outreach Coordinator, TSI Winter 2022 – Summer 2022

Physics Hackathon Judge, McGill Department of Physics 2021 – 2023

STEM Mentor, Fab Fem Organization 2020 – 2021

Wavelength/WaGMiPhAst Co-President, Williams College 2016 – 2020
Student organization supporting marginalized groups in physics & astronomy.

Media & Press Coverage:

Space.com 2025
"James Webb Space Telescope watches our Milky Way galaxy's monster black hole fire out a flare"

Harvard & Smithsonian CfA 2025
"Scientists Make First-Ever Detection of Mid-IR Flares in Sgr A*"

Max Planck Institute for Radio Astronomy 2025
"First-Ever Detection of a Mid-Infrared Flare in Sagittarius A*, the central source of the Milky Way"

Canadian Institute for Theoretical Astrophysics / University of Toronto 2025
"First-Ever Detection of Mid-IR Flares in Sgr A* - the Supermassive Black Hole at the Galactic Center"

Phys.org 2025
"First-ever detection of a mid-infrared flare in Sagittarius A*, the Milky Way's supermassive massive black hole"

Live Science 2025
"James Webb telescope captures 1st 'mid-infrared' flare from Milky Way's supermassive black hole"

NICER / ISS Science Nugget 2025
"Preparing for the Sombrero's shadow"

McGill News 2024
"McGill researchers help uncover rare gamma-ray flare from a distant black hole"